

The open, safety-engineered power-conversion platform

One reusable dual-MCU control module. Four power-stage chassis spanning approximately 50 kW to the megawatt class. Motor drives and bidirectional DC-DC conversion, with IGBTs today and SiC-ready designs. Every schematic, line of firmware, safety analysis, and test trace is published.



The Size 2 control and power stack is built, bench-tested, and undergoing validation.

400 V class

150, 200, or 400 V by DC-link selection

465 A

Continuous · 600 A peak (60 s)

10

Published test reports, raw telemetry linked

PMSM + induction

Instrument-verified motor self-commissioning

■ Built and bench-tested

Size 2 (approximately 100 kW) has completed a **180 V bring-up**, a **one-hour endurance run**, a **20-minute reversal stress test**, and instrument-verified **motor self-commissioning**. Next, a Size 2 chassis is being integrated into our 2018 Zero DSR electric motorcycle.

■ Safety engineered

Dual-MCU architecture with **six redundant safe-state pathways**. The hazard analysis (HARA) and FMEA map to **ISO 26262** and **IEC 61800-5-2**. Every hazard traces to a safety requirement and a test.

■ Open everything

Schematics, firmware, the **RTE Studio** model-based workflow (no MATLAB; **alternative control algorithms benchmarked against FOC**), safety analyses, and **interactive telemetry logs** are all public at docs.openvvf.org.

PLATFORM FAMILY

One control module, four power stages

CONTROL MODULE Dual-MCU · FOC motor control + independent safety coprocessor · USB-B, dual CAN FD, and expansion I/O · RTE Studio + Telemetry Viewer



Size 1

≈50 kW class inverter for light EVs and lab benches

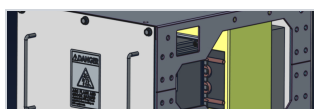
■ **EARLY DESIGN**



Size 2 inverter

≈100 kW · 400 V · 465 A · bench-tested

■ **IN VALIDATION**



Size 2 DC-DC

Bidirectional DC-DC for charging, storage, and microgrids

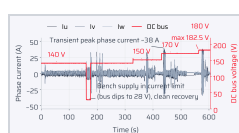
■ **IN PROGRESS**



Size 3

Megawatt class · 900 V · 1,400 A · liquid-cooled

■ **IN DESIGN**



Live telemetry, not promises

Test reports include interactive logs for the browser-based Telemetry Viewer. Scan to open the 180 V bring-up: 140 to 180 V bus steps, approximately 38 A peak current, and clean recovery from a supply current-limit event.



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